TASK 4: DPF/SCR AFTERTREATMENT DIAGNOSIS AND VERIFICATION

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 6 | v1.1

TIME
90 minutes (adapt to workplace conditions)

EVIDENCE
Practical | Knowledge | Documentation

PROGRAMME POINT
Weeks 75-90 | Months 20-24

Your task

Investigate an emissions warning or derate involving DPF/SCR/AdBlue systems. Use data and physical checks to find the cause and decide whether regeneration, repair or referral is appropriate.

What you must do

1. Confirm warning history, operating pattern, fluid quality and relevant DTCs.
2. Inspect exhaust, dosing, sensor and wiring condition safely.
3. Record temperatures, pressures, NOx or dosing data against specifications.
4. Determine whether prerequisites permit an authorised regeneration or functional test.
5. Identify root cause, complete permitted action and verify emissions-system status.

What good evidence looks like

SAFE CONTROL
Regeneration requires an approved exclusion zone, ventilation and continuous monitoring; hot exhaust and fire risks must remain controlled.

CORRECT METHOD
A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE
Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS
A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle or aftertreatment rig
- Temperature/pressure and fluid-quality test equipment
- OEM-capable scan tool
- Technical data, exclusion controls and PPE

SOW ALIGNMENT

Block 6 - Engines, Fuel, Emissions, Thermal Management and Emerging Technology

KSB evidence focus: S2, S4, B3; reinforced K1-K4, K6, K10, S1, S3, S5-S6, S8-S10, B2

Task reference: HV_L3_Block_6_Task_4_DPF_SCR_Aftertreatment_Diagnosis_and_Verification

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

DTC / warning state		DPF differential pressure	
Exhaust temperatures		Soot / ash estimate	
AdBlue quality / level		NOx / dosing result	
Regeneration prerequisites		Outcome / verification	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why must the root cause be addressed before repeated regeneration?

2. What is the difference between soot loading and ash accumulat

Knowledge check - continued

3. Explain the hazards of stationary regeneration.

4. How can poor operating pattern influence aftertreatment condition?

5. What evidence confirms repair or regeneration success?

Assessor/Workplace Mentor observation

- ☐ Establishes controlled area and exhaust/fire precautions.
- ☐ Assesses regeneration prerequisites and authority.
- ☐ Records outcome, resets and verification appropriately.
- ☐ Uses DTCs, data and physical checks together.
- ☐ Identifies system cause rather than treating one symptom.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

☐ PASS ☐ REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

FAULT SCENARIOS AND EVIDENCE

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 6 | Task 4

How to use this sheet

Work through the listed scenarios during the block using a vehicle, safe fault rig or supported case study. Record which scenario was used for this task and reference the actual measurements and assessor observation on the existing practical record. A discussion or case study alone is not evidence of practical competence.

1. SCR fault

Checks: Confirm derate/warning and relevant DTCs; check AdBlue quality/supply, dosing, NOx sensors, exhaust leaks and prerequisites.

Evidence: Record data and permitted functional-test results; verify system status without clearing evidence prematurely.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

Scenario selected for the original task assessment: _____

For any safety-critical fault, stop or control the work and follow the manufacturer, DVSA and workplace procedures. Use the original task pages for findings, feedback, PASS/REFER and signatures.

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.